MINOR PROJECT REPORT

Predictive Classification using Random Forest Ensembles

AIML Minor Project 4

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Dataset	Breast Cancer Wisconsin (Diagnostic)
Algorithm	RandomForestClassifier
Submission date	11 October 2026

Educational classification project. This model is not intended for clinical diagnosis or medical decision-making.

Abstract

This project implements a Random Forest ensemble classifier on the Breast Cancer Wisconsin (Diagnostic) dataset. The dataset contains 569 observations and 30 numeric predictor features. The target label is diagnosis: B (benign) or M (malignant). A stratified 80:20 train-test split was used. Missing-value handling is included in a preprocessing pipeline, although the included dataset contains 0 missing values. The model achieved accuracy 0.965, malignant-class precision 1.000, malignant-class recall 0.905, malignant-class F1-score 0.950, and ROC-AUC 0.996 on the held-out test set.

1. Introduction

Classification is a supervised machine-learning task in which a model learns patterns from labeled examples and predicts a discrete class for unseen records. Random Forest is an ensemble method that combines many decision trees. Each tree is trained using bootstrapped samples and randomized feature subsets; the trees vote to produce the final class. The ensemble can reduce variance compared with a single decision tree and can model nonlinear relationships.

2. Project Objectives

• Explore the data, class distribution, feature distributions, and correlations.

• Build a repeatable preprocessing pipeline and use an 80:20 train-test split.

• Train a Random Forest classifier and justify the selected hyperparameters.

• Evaluate performance using confusion matrix, precision, recall, F1-score, and ROC-AUC.

• Visualize feature importance and document limitations and possible improvements.

3. Dataset Description

The dataset is the Breast Cancer Wisconsin (Diagnostic) benchmark, available on Kaggle and originating from the Wisconsin Diagnostic dataset maintained by UCI. The included CSV uses scikit-learn's bundled copy of the same benchmark with 30 numeric predictor features and the diagnosis label. Some Kaggle versions include an ID column and an extra empty column; those are not included because they are not useful predictors. The data contains 357 benign cases (62.7%) and 212 malignant cases (37.3%).

Target labels: B = benign (negative class); M = malignant (positive class). The malignant class is treated as positive for precision, recall, F1, ROC curve, and confusion-matrix interpretation.

Dataset sources: Kaggle — https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data ; UCI — https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

4. Exploratory Data Analysis (EDA)

The dataset contains 569 rows, 30 predictor columns and one target column. All predictor features are numeric measurements derived from digitized cell nuclei images, including radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. The class distribution is moderately imbalanced: benign cases outnumber malignant cases. This makes it important to examine recall and F1-score in addition to accuracy.

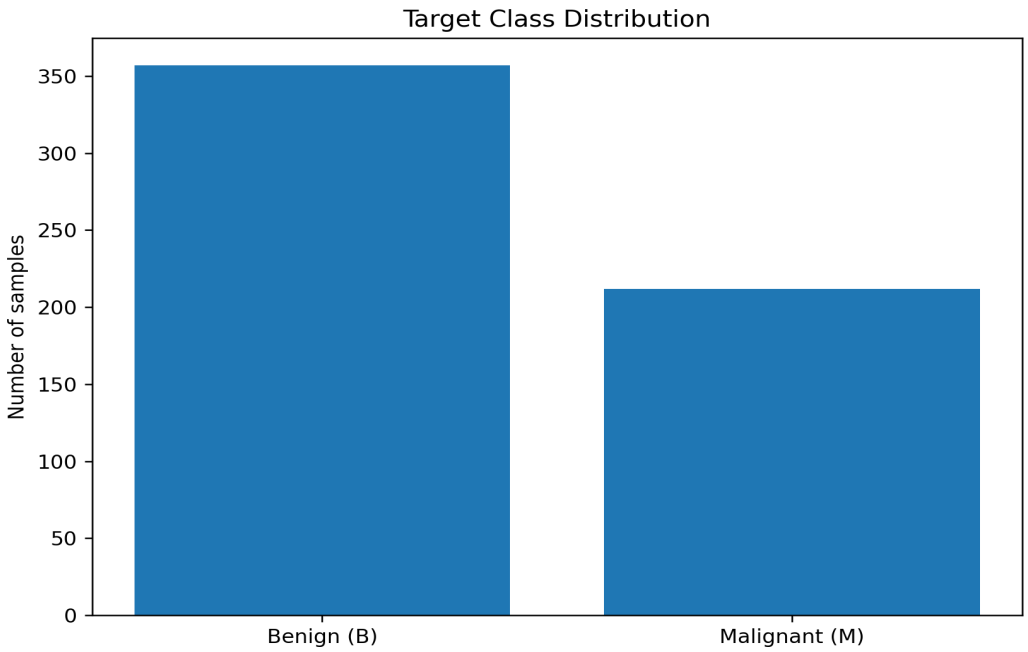


Figure 1. Distribution of the two diagnosis classes.

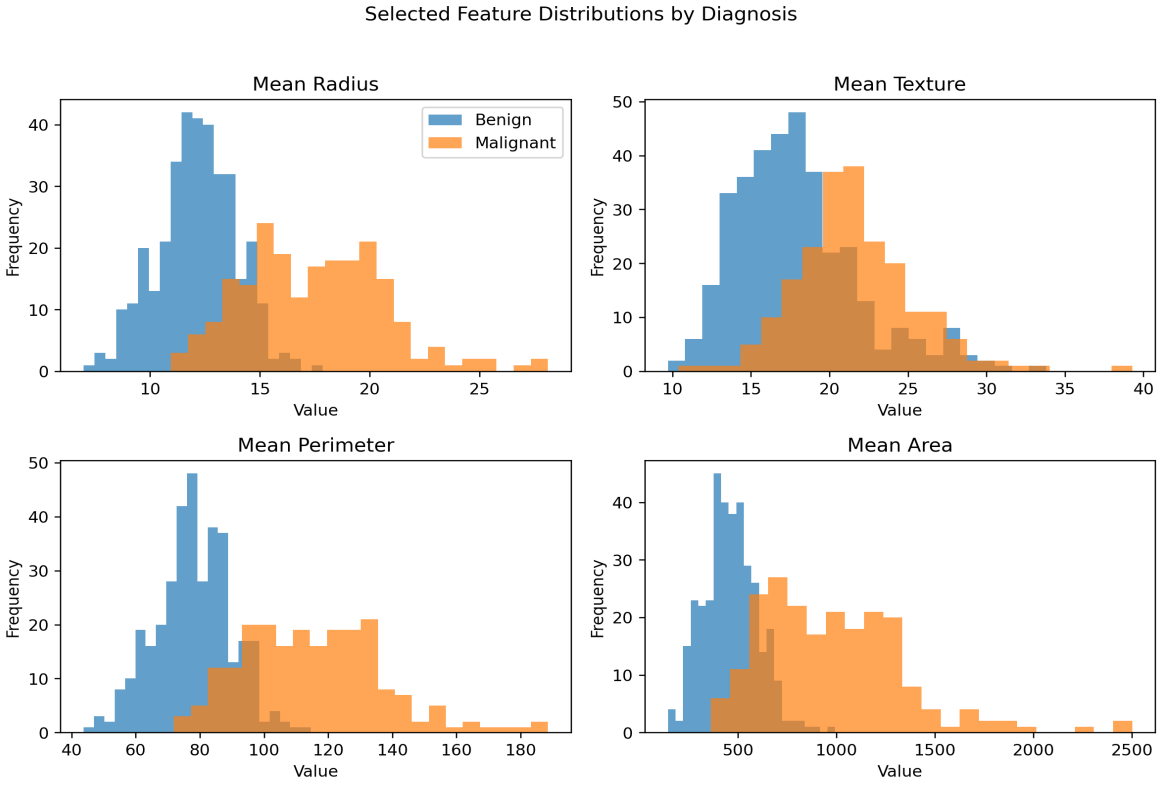


Figure 2. Selected feature distributions by diagnosis.

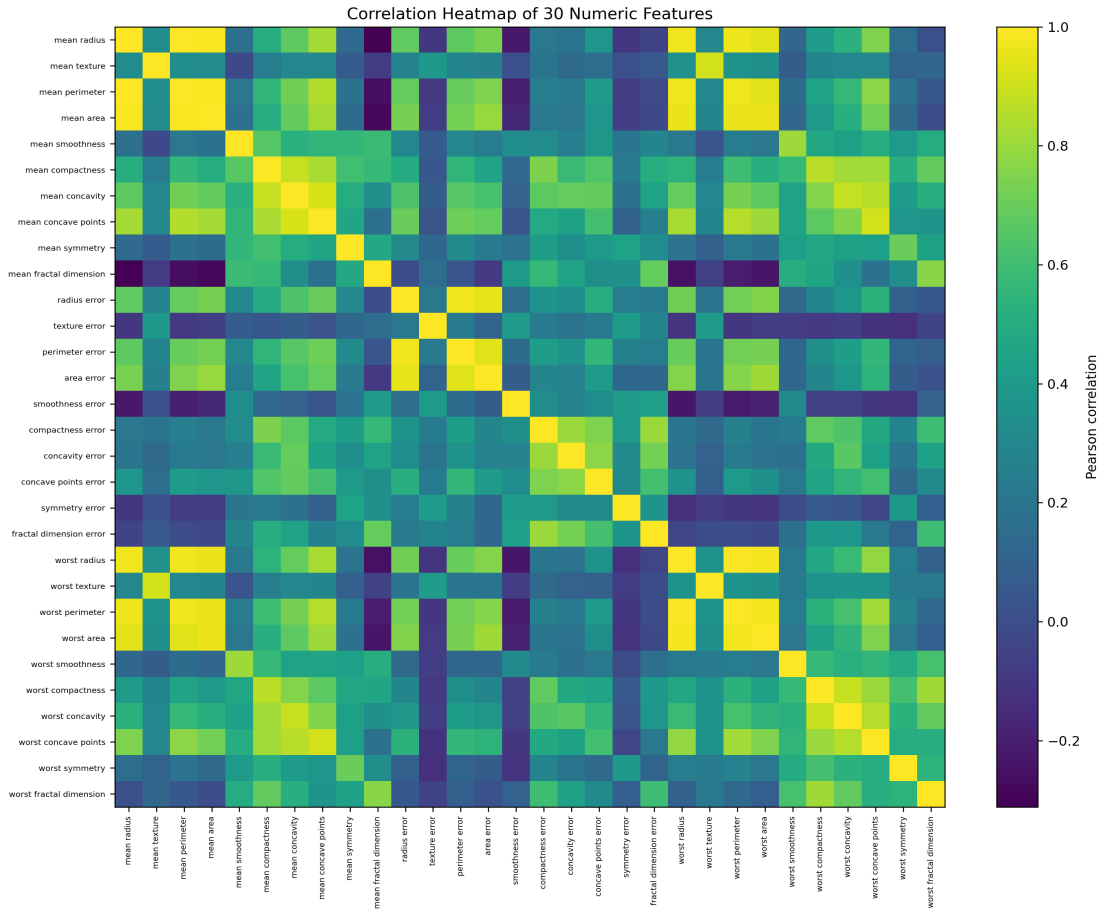


Figure 3. Correlation heatmap for the numeric predictor features. Strong correlations indicate overlapping information; correlation does not prove causation.

5. Data Preprocessing

The target column is diagnosis. Predictor columns are all numeric, so no categorical predictor encoding is required for this selected dataset. The training script includes a median imputer for numeric values and supports most-frequent imputation plus one-hot encoding if categorical columns are added in a future dataset. The included data has no missing values; the missing-value report is still generated to document this check.

The dataset is split into 80% training data and 20% testing data using stratification, which preserves class proportions. The random seed is 42 for reproducibility. Feature scaling is not applied because Random Forest is based on decision-tree splits and does not depend on Euclidean distance or gradient magnitudes.

6. Model Development

The selected model is RandomForestClassifier with n_estimators=200, max_depth=10, class_weight='balanced', and random_state=42.

• n_estimators = 200: a moderately large forest helps stabilize the ensemble vote while remaining practical to train.

• max_depth = 10: limiting depth controls model complexity and helps reduce overfitting.

• class_weight = 'balanced': adjusts weights inversely to class frequencies to reduce bias toward the majority class.

• random_state = 42: supports repeatable results.

7. Evaluation Results

The test set contains 114 observations. Metrics below are calculated only on this held-out set, not on the training set.

Metric	Result	Interpretation
Accuracy	0.965	96.5% of the held-out test records were classified correctly.
Precision (malignant)	1.000	Of cases predicted malignant, this fraction were actually malignant.
Recall (malignant)	0.905	Of all actual malignant cases, this fraction were identified.
F1-score (malignant)	0.950	Harmonic mean of malignant precision and recall.
ROC-AUC (malignant)	0.996	Discrimination ability across probability thresholds; 1.0 is ideal.

7.1 Confusion Matrix

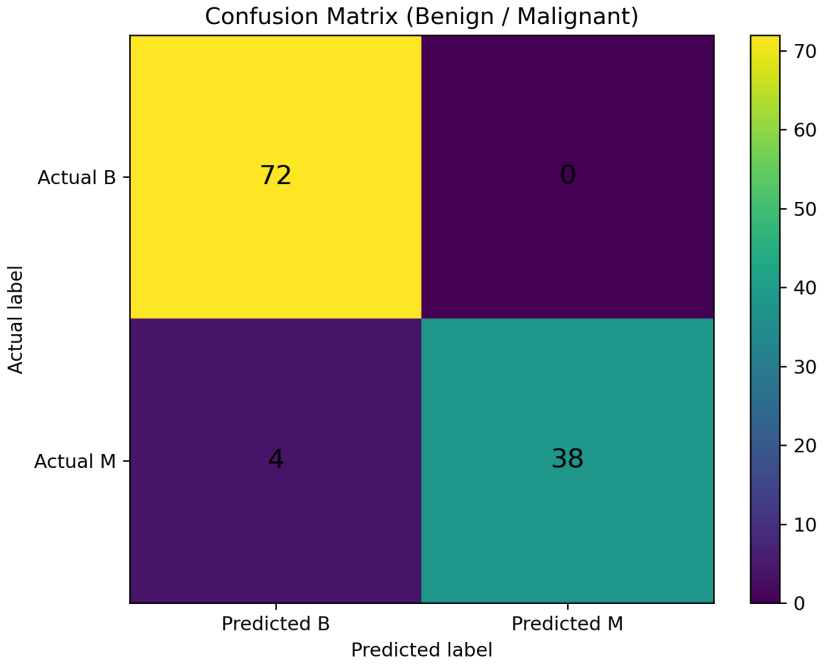


Figure 4. Confusion matrix with benign and malignant labels.

	Predicted benign (B)	Predicted malignant (M)
Actual benign (B)	72	0
Actual malignant (M)	4	38

For malignant (M) as the positive class: TN=72, FP=0, FN=4, and TP=38. True negatives are benign cases correctly predicted benign. False positives are benign cases incorrectly predicted malignant. False negatives are malignant cases incorrectly predicted benign; in a screening context, false negatives are especially important to minimize. True positives are malignant cases correctly identified.

7.2 Precision, Recall, and F1-score

Malignant precision is 1.000. Among records predicted malignant, this fraction were actually malignant. Malignant recall is 0.905; among actual malignant records, this fraction were identified. F1-score is 0.950, the harmonic mean of precision and recall. The appropriate balance depends on the use case; this coursework model should not be used to guide clinical decisions.

7.3 ROC Curve and AUC

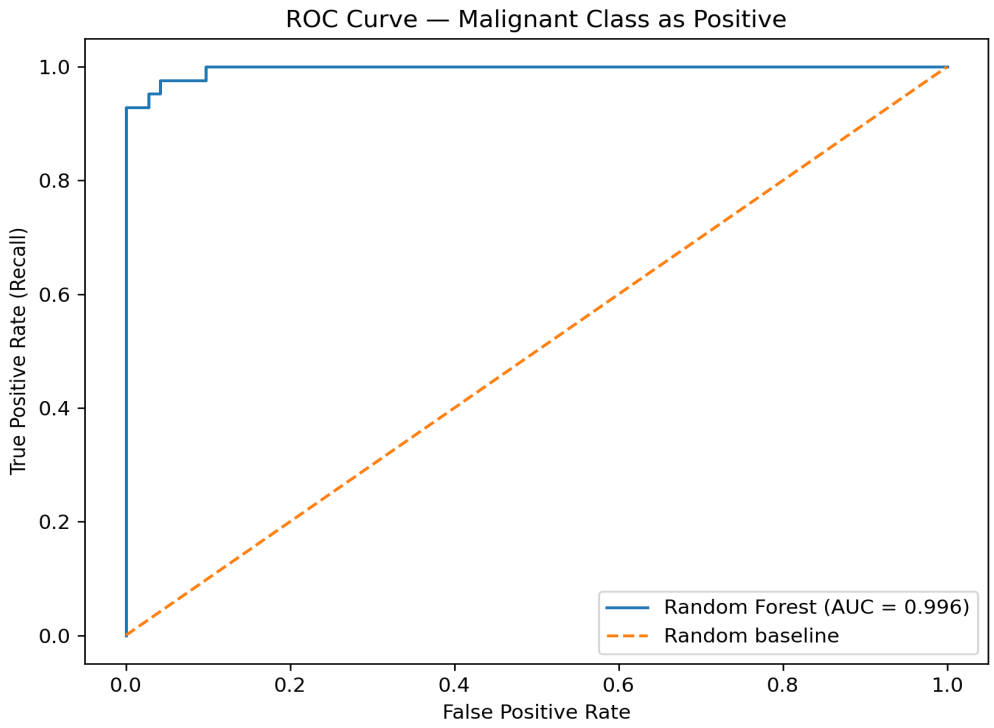


Figure 5. ROC curve for malignant diagnosis as the positive class.

The ROC curve plots true-positive rate against false-positive rate at different probability thresholds. The measured AUC is 0.996. A value closer to 1 indicates stronger ranking discrimination, while 0.5 is approximately random ranking.

7.4 Feature Importance

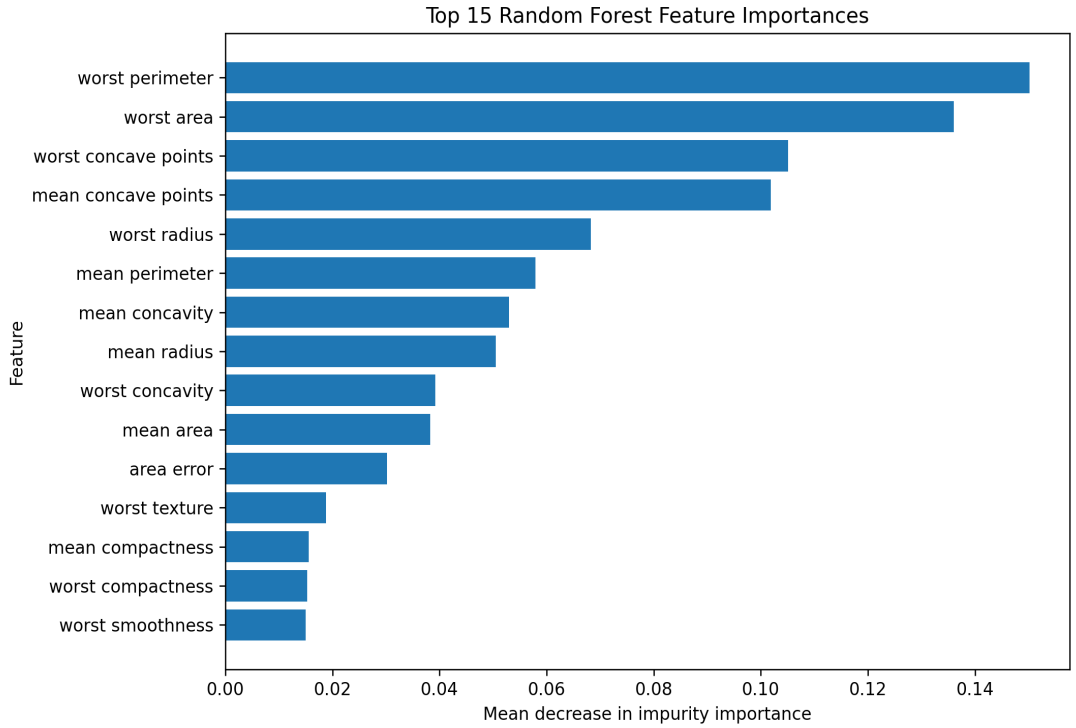


Figure 6. Top 15 feature importances from the fitted Random Forest.

The highest-ranked features in this fitted model are: worst perimeter (0.150), worst area (0.136), worst concave points (0.105), mean concave points (0.102), worst radius (0.068), mean perimeter (0.058), mean concavity (0.053), mean radius (0.051). Feature importance reflects how the fitted trees used features to reduce impurity. It does not establish causation, and correlated predictors can share or distort importance.

8. Strengths and Limitations

Strengths: Random Forest can model nonlinear feature relationships, combines many trees to reduce variance, supports feature-importance summaries, and works without feature scaling. The pipeline is reproducible and includes missing-value handling.

Limitations: the result depends on one random 80:20 split; the dataset is relatively small and moderately imbalanced; feature importance can be biased when features are correlated; and the model is not clinically validated. Metrics from this benchmark do not establish performance on new hospitals, populations, or imaging equipment.

9. Future Improvements

Potential next steps include repeated stratified cross-validation, hyperparameter search over n_estimators/max_depth/min_samples_leaf/max_features, comparison with Gradient Boosting and Logistic Regression, threshold tuning based on the cost of false negatives and false positives, calibration analysis, and external validation on independent data. Any real medical use would require independent clinical validation and expert oversight.

10. Conclusion

The project demonstrates a complete classification workflow from EDA and preprocessing to Random Forest training and evaluation. On the held-out test split, the model achieved 0.965 accuracy, 1.000 malignant precision, 0.905 malignant recall, 0.950 malignant F1-score, and 0.996 ROC-AUC. These results should be interpreted cautiously and are not evidence of clinical readiness.

References

• Kaggle. Breast Cancer Wisconsin (Diagnostic) Data Set. https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

• UCI Machine Learning Repository. Breast Cancer Wisconsin (Diagnostic).
https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

• scikit-learn documentation. Breast cancer dataset loader.
https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

• scikit-learn documentation. RandomForestClassifier.
https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

Appendix: Reproducibility

Run `pip install -r requirements.txt`, then `python train_model.py`. The training script recreates metrics, plots, the classification report, test predictions, and the saved model. Run `streamlit run app.py` to open the dashboard.